

Efficient Micro-Expression Spotting Based on Main Directional Mean Optical Flow Feature

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ABSTRACT

Human facial expressions can convey a great deal of information in daily life. Spotting macro-expression (MaE) and micro-expression (ME) intervals from long video sequences is a difficult challenge. In this paper, we propose an efficient framework for the expression spotting task. This framework consists of three main modules: Face Cropping and Alignment Module (FCAM), optical flow Feature Extraction Module (FEM), and expression Proposal Generation Module (PGM). The noise of optical flow features is reduced by face cropping and alignment, and the Main Directional Mean Optical Flow Feature of the regions of interest is extracted as the feature for expression spotting. Finally, the expression intervals are spotted by our designed expression proposal generation module. Our approach achieves very good results on the SAMM Long Videos and CAS(ME)². To demonstrate the transferability of our method, we tested it on the MEGC2023 unseen dataset and finally achieved the third place, proving the effectiveness of our method.

CCS CONCEPTS

• Computing methodologies → Computer vision tasks.

KEYWORDS

Optical Flow, Expression Spotting, Micro expression, Macro expression

ACM Reference Format:

Jun Yu, Zhongpeng Cai, Shenshen Du, Xiaxin Shen, Lei Wang, and Fang Gao. 2023. Efficient Micro-Expression Spotting Based on Main Directional Mean Optical Flow Feature. In *Proceedings of the 31st ACM International Conference*

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MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

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ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612861>

on Multimedia (MM '23), October 29–November 3, 2023, Ottawa, ON, Canada.
ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612861>

1 INTRODUCTION

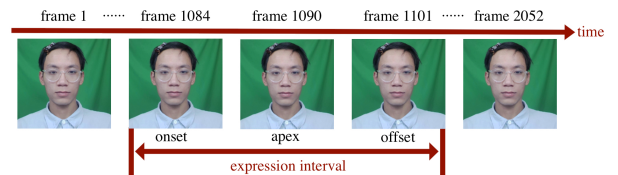


Figure 1: Macro- and micro-expression spotting task process.

Facial micro-expressions are involuntary, fleeting facial expressions that occur spontaneously in response to an emotion but are immediately suppressed. These minute expressions, lasting for less than half a second, reveal fleeting glimpses of a person's true emotions, even when those emotions are being actively concealed. Although the boundary between macro- and micro-expressions is blurred, researchers usually distinguish micro-expressions from macro-expressions by the duration and intensity of facial expressions, and research scholars generally believe that micro-expressions with a duration of less than 0.5 s and small amplitude of facial muscle movement [16].

The goal of the task of face expression spotting is to find the onset and offset frames of expression occurrence in a long video sequence. The processing of the macro-expression micro-expression spotting task is shown in Figure 1. Some traditional methods [11, 24, 25] have achieved good results on micro-expression spotting tasks. In recent years, many deep learning methods [7, 9, 14] have begun to dominate with the presentation of large micro-expression datasets. Among them, the emergence of CAS(ME)² [13], CAS(ME)³ [7], SAMM Long Videos (SAMM-LV) [19], SMIC-E-long [14], SAMM [1] and 4DME [9] led a wave in the development of micro-expression technology.

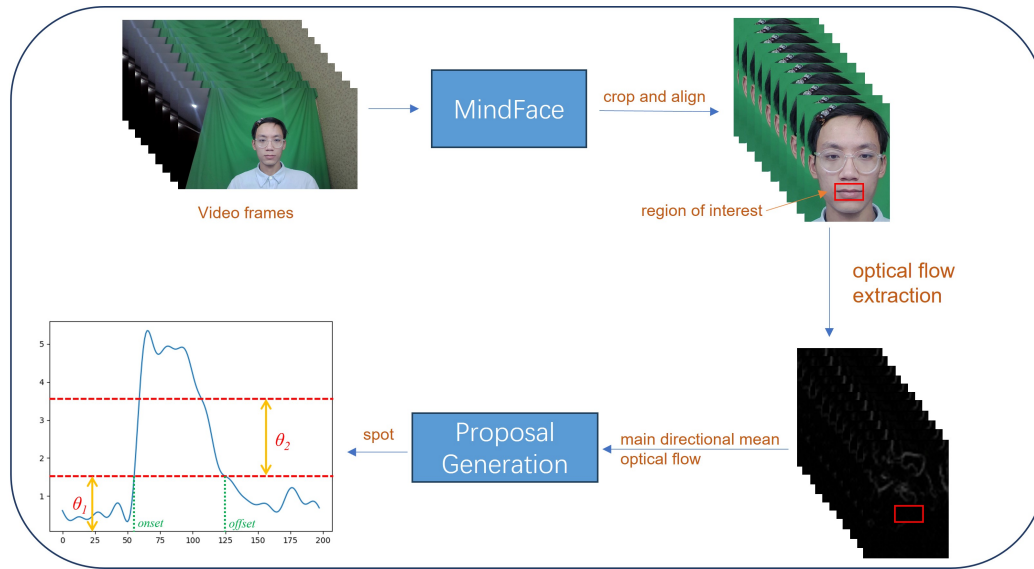


Figure 2: The pipeline of our proposed method.

In this article, we presented our approach to the Micro-Expression Grand Challenge. We first used the face detection function in the MindFace [10] toolkit to extract the faces from the video frames. Then, we use Dlib [5] to align these faces. After that, we selected five regions of interest from the cropped facial parts and extracted the optical flow features using the Main Directional Mean Optical flow feature (MDMO) method [21]. Finally, we spotted the final facial expression segment through an Expression Proposal Generation module.

To sum up, our contributions can be summarized as:

- We proposed a complete and universal method for macro and micro-expression spotting task, which can efficiently spot the onset and offset frames of the expressions from long videos.
- We used the Main Directional Mean Optical flow feature as the expression feature and proposed a proposal generation module (PGM) to accurately spot the expression intervals.
- Our method achieved good results in the MEGC2023 spotting task, with a third-place finish on the unseen dataset, further demonstrating the effectiveness of our proposed method.

2 RELATED WORK

2.1 Micro expression spotting method

Moilanen et al. [11] used a Local Binary Pattern (LBP) algorithm to extract facial features, and then utilized the feature differences to find vertex frames. Li et al. [8] extracted features from projections in PCA space and used SVM to distinguish ME from other facial movements. Pan et al. [12] proposed a local bilinear convolutional neural network based on fusing global and local features of faces in each frame of a long video sequence with good results. Wang et al. [15] proposed MSE network to extract the temporal and spatial features of video through a 2+1D spatio-temporal convolutional network achieved good results. Yap et al. [20] proposed a

3D-CNN network to extract spatio-temporal features by comparing each frame with two temporally localized reference frames via a convolutional model.

2.2 Micro-Expression Grand Challenge

In MEGC2021, Yang et al. [17] designed an end-to-end deep learning framework for micro-expression spotting by using the detected facial action units as feature quantities. Yu et al. [23] migrated the method of action localization to micro-expression spotting, and greatly improved the performance of the deep learning method for micro-expression spotting task by introducing the feature pyramid structure and the self-attention mechanism. He et al. [24] worked through very detailed data processing designed a method based on localized optical flow in 14 regions of interest of the face, which is far more effective than general deep learning methods. In MEGC2022, most of the participants followed He's method and made improvements based on it. Yu et al. [22] improved the precision score by introducing an EMD [4] method for secondary screening of expression candidate segments. Zhao et al. [26] achieved good results by constructing a mixture of Bayesian optimization and Nash equilibrium to search for the optimal parameters. Leng et al. [6] proposed the Apex and Boundary Perception Network (ABPN) to generate expression candidate segments by extracting features from optical flow features using a CNN.

3 METHODOLOGY

As shown in figure 2, our proposed method mainly consists of three main modules: Face Cropping and Alignment Module (FCAM), optical flow Feature Extraction Module (FEM), and expression Proposal Generation Module (PGM).

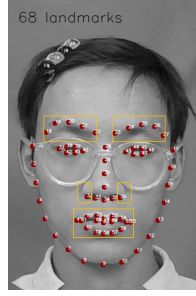


Figure 3: Facial landmarks and Regions of Interest.

3.1 Face Cropping and Alignment Module

Throughout the methodologies that have utilized optical flow as the feature, face cropping and alignment are crucial. Because, without proper face cropping, background information in video frames can cause significant interference. Moreover, if the alignment of the face is not accurate, the extracted optical flow features can be very noisy. We first used Mindface [10] to extract faces from video frames and perform cropping. Mindface uses RetinaFace [2] as its detection approach, which is efficient in detecting faces. Then, we used Dlib [5] to detect 68 facial landmarks, and aligned all faces in the same video based on landmarks 31, 40, and 43, the specific location of the 68 facial landmarks can be referred to Figure 3.

Despite the face alignment we did, there was a slight movement of the face. So we have to do one more alignment of the optical flow features. We first compute the main directional mean optical flow feature $\bar{\rho}_{nose}$ of the nose tip region, then we consider the global motion optical flow of the face $\bar{\rho}_{face} = \bar{\rho}_{nose}$ and subtract this global optical flow in all subsequent optical flow computations of the region of interest.

3.2 Feature Extraction Module

According to the Facial Coding System [3], facial expressions can be decomposed into various components of muscle movement. Therefore, we can choose the movements of certain local facial regions to represent the movements of the entire face. We have selected the movements of two eyebrows, both nostrils, and the mouth area as the regions of interest for extracting optical flow features, and used the features from these five areas to represent the facial movements in generating expressions. Our definition of face landmarks and the selection of ROIs are shown in Figure 3.

We first used the Gunnar Farneback method to compute the dense optical flow vectors $\vec{V}^i = [\vec{V}_x^i, \vec{V}_y^i]$ for each point within the regions of interest. Here, $\vec{V}_x^i$ represents the optical flow in the x-direction, and $\vec{V}_y^i$ represents the optical flow in the y-direction, where $i = 1, 2, \dots, 5$ is the index of ROIs. We then calculate the Main Directional Mean Optical Flow Feature ρ_i within each interest using the following equation:

$$\rho_i = \frac{1}{S_i} \sqrt{\vec{V}_x^i{}^2 + \vec{V}_y^i{}^2} \quad (1)$$

Where, S_i is the area of the i th region of interest. Then, we subtract the global motion features of the face to get the final optical flow

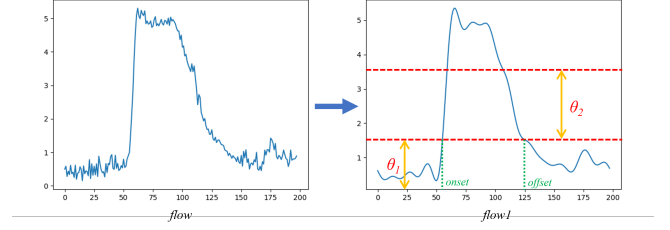


Figure 4: Expressions Proposal Generation Module Process.

features $f_i = \rho_i - \rho_{face}$ for each region of interest. Finally, we obtained the features of each video frame as $f^t = [f_1, f_2, f_3, f_4, f_5]$, where $t = 2, 3, \dots, N$ is the index of frames.

3.3 Proposal Generation Module

The flow of the expression proposal generation module is specifically shown in Figure 4. We plotted the optical flow features $flow$ of each ROI region as a curve with the number of frames on the horizontal axis and the intensity of the optical flow on the vertical axis. Considering that our extracted optical flow features are still very noisy, we designed a low-pass filter to filter out the high-frequency noise. In this way we obtain a clean optical flow features $flow_1$. Then we went to filter the segments in this curve whose intensity exceeded a threshold θ_1 , which expresses that the region produced a certain amount of action, and then for these segments, we further filtered the segments whose amplitude difference was greater than θ_2 , which indicates that the action reached a certain intensity and time. Finally, we do a fusion based on the intersection ratio of these five regionally localized clips, and if the overlap ratio reaches α , we consider that this is a movement experience of an expression in different regions of the human face, and we need to do a fusion to get the final result.

4 EXPERIMENTS

4.1 Datasets

We chose the CAS(ME)² [13] and SAMM-LV [19] to verify the performance of our method. These two datasets are classic datasets for micro-expression spotting tasks, and the test results on these datasets can reflect the performance and generalization of our method. There are 22 subjects and 98 long videos, and the average duration is 148s. The samples contain 300 MaEs and 57 MEs in total. In SAMM-LV, there are 147 videos which contain 343 MaEs and 159 MEs in total.

Furthermore, the unseen testing set (MEGC2023-TestSet) is used to evaluate the performance of our method. It contains 30 long video, including 10 long videos from SAMM [1] (SAMM Challenge dataset) and 20 clips from different videos in CAS(ME)³ [7]. The frame rate for SAMM is 200 fps and the frame rate for CAS(ME)³ is 30 fps.

4.2 Setup

For face cropping, we used the RetinaFace-mobilenet025 model pre-trained on WiderFace [18]. The face images in the SAMM are

Table 1: The leaderboard of ME and MaE spotting task.

Team	Both			SAMM			CAS(ME) ³		
	F1 Score	Precision Score	Recall Score	F1 Score	Precision Score	Recall Score	F1 Score	Precision Score	Recall Score
xuke	0.22	0.28	0.19	0.37	0.37	0.36	0.21	0.27	0.17
qinwf	0.19	0.25	0.16	0.29	0.29	0.29	0.18	0.24	0.15
Ours	0.18	0.18	0.19	0.31	0.28	0.35	0.17	0.16	0.17
Kuchen	0.14	0.12	0.17	0.29	0.27	0.31	0.12	0.10	0.16
jyy	0.12	0.20	0.09	0.27	0.29	0.25	0.10	0.18	0.07
panhang	0.10	0.15	0.08	0.16	0.13	0.20	0.09	0.16	0.06
yumemiso	0.10	0.10	0.09	0.03	0.07	0.02	0.10	0.10	0.10
gaozhenwei	0.10	0.15	0.07	0.07	0.07	0.07	0.10	0.18	0.07

resized to 320×320 and the faces in the CAS(ME)³ are resized to 256×256. For SAMM, θ_1 is set to 1.4, θ_2 is set to 1.5. For CAS(ME)³, θ_1 is set to 1.65, θ_2 is set to 1.8. Finally, the IoU threshold α is set to 0.2, and whenever the IoU of two expression intervals exceeds 0.2, then the two intervals are merged. Experiments are also conducted using MindSpore.

4.3 Metrics

Given a set of predicted expression intervals $\hat{Y}$ and a set of ground truth expression intervals Y , we calculate the IOU between each predicted interval and each ground truth interval using the formula:

$$IOU(\hat{y}, y) = \frac{|\hat{y} \cap y|}{|\hat{y} \cup y|} \quad (2)$$

where $\hat{y}$ and y represent the predicted and ground truth intervals, respectively.

We then determine the maximum IOU value for each predicted interval and compare it to the threshold k . If the maximum IOU value is greater than or equal to the threshold, we consider the predicted interval to be a TP. Formally, if we define the set of TP as:

$$TP = \hat{y} \in \hat{Y} : \max_{y \in Y} IOU(\hat{y}, y) \geq k \quad (3)$$

where k is set to 0.5.

Precision refers to the probability of correctly identifying the positive samples among all samples predicted as positive. It can be calculated using the formula:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

Recall, refers to the probability of correctly predicting the positive samples among all actual positive samples. Recall can be calculated using the formula:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

For the expression spotting task, we naturally want both precision and recall to be high. Therefore, we also use the F1 score to evaluate the performance of our method. The F1 score is the harmonic mean of precision and recall, calculated using the following formula:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

By computing the precision, recall and F1 score, we can evaluate the performance of our method.

4.4 Results

We first report the results of our experiments on CAS(ME)² and SAMM-LV. We compare our results with some state-of-the-art methods, the specific experimental results are shown in Table 2. It can be seen that the performance of our method is greatly improved on both micro-expression and macro-expression spotting tasks. The fact that the results are good on both datasets also indicates that the transferability of our method is good.

Table 2: F1 score comparison with the state-of-the-art models on CAS(ME)² and SAMM-LV.

Method	CAS(ME) ²			SAMM-LV		
	MaE	ME	ALL	MaE	ME	ALL
Baseline [25]	0.2131	0.0547	0.1403	0.0725	0.1331	0.0999
Yang et al. [17]	0.2599	0.0339	0.2118	0.3553	0.1155	0.2736
Yu et al. [23]	0.3800	0.063	0.327	0.3360	0.2180	0.2900
He et al. [24]	0.3782	0.1965	0.3436	0.4149	0.2162	0.3638
Leng et al. [6]	0.3357	0.1590	0.3117	0.3349	0.1689	0.2908
Ours	0.4260	0.4185	0.4000	0.4168	0.2240	0.3680

To better demonstrate the versatility of our approach, we report the rankings on the MEGC2023 unseen test set in the Table 1. We finally ranked third, proving the effectiveness of our method. It is worth mentioning that our method has the highest recall among all the teams, indicating that our method still has a lot of potential.

5 CONCLUSION

In this paper, we present an efficient method for expression spotting task. We divide the face expression spotting task into three modules: face cropping and alignment, optical flow feature extraction, and expression proposal generation. By optimizing each module, our method achieves good results in macro-expression and micro-expression spotting task. In MEGC2023, our method achieves the third place, which proves the effectiveness of our approach.

ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of China (62276242), National Aviation Science Foundation (2022Z071078001), CAAI-Huawei MindSpore Open Fund (CAAIJSJLJJ-2021-016B, CAAIJSJLJJ-2022-001A), Anhui Province Key Research and Development Program (202104a05020007), USTC-IAT Application Sci. & Tech. Achievement Cultivation Program (JL06521001Y), Sci. & Tech. Innovation Special Zone (20-163-14-LZ-001-004-01).

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